

REVIEWER REPORT

The manuscript addresses an ambitious and potentially interesting problem at the interface of nonlinear dynamics, transfer-operator methods, random-matrix ideas, and the Riemann zeros. The author is careful in several places to state that the numerical observations do not constitute a proof of the Riemann Hypothesis. However, the central claims of the paper are not supported by a sufficiently well-defined mathematical operator, an independent validation strategy, or internally consistent numerical evidence. The current results are better interpreted as a target-calibrated numerical reconstruction than as evidence of a spectral isomorphism.

The deficiencies are structural rather than editorial. They affect the definition of the object whose spectrum is studied, the derivation of the cooling law, the use of the Riemann zeros in parameter selection, the interpretation of the conjugate-spectrum fit, and the reproducibility of the reported numerical values. For this reason, I do not believe that the manuscript can be made publishable through a conventional major revision. A substantially redesigned study would be required.

Major Comments

1. Meaning of “spectral isomorphism” and the Hilbert-Pólya connection

The title makes a strong claim of spectral isomorphism. Yet the manuscript first invokes a unitary evolution operator U on a Hilbert space and then computes the eigenvalues of an empirical real transition matrix P . The construction of U , its unitarity, and its precise relation to P are not supplied. Moreover, the Riemann-zero ordinates are compared with rescaled eigenphases rather than being obtained as eigenvalues of a self-adjoint operator. The manuscript therefore does not establish a Hilbert-Pólya-type spectral realization. At minimum, the two spectral objects and the exact meaning of “isomorphism” would need a formal mathematical definition and proof-level justification.

2. The proposed $1/\log^2(n)$ law is heuristic, not a rigorous derivation

The argument combines the Riemann-von Mangoldt mean-spacing behavior with an asserted square-root relation between parameter perturbation and eigenphase spacing, and then imposes a heuristic proportionality between the latter and the Riemann-zero spacing. The manuscript itself labels one of these steps a heuristic correspondence. Consequently, statements that the $1/\log^2(n)$ law is strictly dictated, exact, or that alternative logarithmic forms are fundamentally invalid are not justified by the derivation presented.

3. Target leakage and lack of genuine out-of-sample prediction

The most serious numerical issue is that the Riemann zeros are used repeatedly to determine the numerical and dynamical parameters. In the microscopic regime, ε is selected by minimizing the sum of the absolute errors of the first six predicted levels. The fine scan then identifies $\varepsilon=0.001916$ as the best target-matching value. In the macroscopic regime, k_1 and k_2 are optimized using Differential Evolution against the Riemann spectrum. Since the target data determine the model settings, agreement with the same target cannot establish predictive spectral correspondence. A credible validation would require fixing all model choices on a clearly specified training set and then testing, without retuning, on a pre-declared unseen range of zeros.

4. Non-autonomous transfer operator and normalization of P

For a non-autonomous dynamical system, a time-ordered product/cocycle or an explicitly defined non-autonomous transfer-operator framework is normally required. The manuscript instead diagonalizes an arithmetic time average of transition matrices and interprets that spectrum as the dynamical spectrum without providing a theorem or numerical convergence argument that justifies this substitution. In addition, the Monte Carlo expression $P_{ij}=(1/T)\sum (x_t \text{ in bin } i, x_{t+1} \text{ in bin } j)$ is a joint transition frequency. Unless it is divided by the number of visits to source bin i , its row sums are not generally equal to one, so it is not a conventional row-stochastic transition matrix. This point must be resolved because it directly affects the computed eigenvalues and eigenphases.

5. Interpretation of the conjugate full-spectrum fit

Strategy D is presented as showing that restoration of $\pm\theta$ pairs makes the fitted ground-state intercept spontaneously approach zero. However, if the regression data are symmetrized as (θ_i, y_i) together with $(-\theta_i, -y_i)$, then both sample means are exactly zero and the ordinary least-squares intercept is necessarily $b=0$. In that case the result is a direct algebraic consequence of constructing a symmetric data set, not evidence for a new physical cancellation mechanism. The manuscript should explicitly list the regression pairs and analytically separate what is enforced by symmetry from what is genuinely learned from the dynamics.

6. Material inconsistency between Table 1, Figure 4, and Figure 5

The manuscript reports different parameter values and MSEs for the same strategies. Table 1 gives approximately $(k_1, \text{MSE}) = (6.761, 21.849)$, $(6.481, 8.479)$, and $(4.390, 9.467)$ for Strategies A, C, and D, respectively. The Figure 4 caption instead reports approximately $(7.429, 20.22)$, $(8.070, 10.08)$, and $(8.783, 10.81)$, and Figure 5 subsequently uses $k_1=8.070$ for Model C. These discrepancies are too large to be explained by rounding and raise a fundamental reproducibility issue.

7. Contradictory description of the large-N scaling protocol

The $N=1000$ Results section characterizes Models M1 and M2 as being evaluated under single-point anchoring using only the first zero and explicitly states that no global fitting is used. The Appendix, however, states that for large data sets a cluster of low-frequency anchor points is used to obtain a globally optimized scaling factor. In addition, the dynamical parameters themselves are globally optimized. These are methodologically different procedures and must not be presented interchangeably.

8. The GUE baseline does not provide a like-for-like test

The paper compares a globally rescaled GUE surrogate with the macroscopic Riemann counting trend and obtains a large MSE. However, the established GUE connection concerns unfolded local statistics such as spacing distributions and pair correlation, not a claim that a single globally scaled semicircle spectrum reproduces the logarithmic mean density of Riemann zeros. The manuscript later acknowledges this distinction. A meaningful RMT comparison should therefore use unfolded nearest-neighbor spacing, pair correlation, number variance, spectral rigidity, or related local-statistical diagnostics.

9. Post-hoc comparison with ion-trap error features

The claimed coincidence of residual peaks near $N \approx 20$ (and later near approximately 20, 40, and 76) is explicitly described as qualitative, which is appropriate. Nevertheless, the manuscript repeatedly uses this visual coincidence to motivate a possible shared resolution-limit mechanism. No pre-specified anomaly statistic, null distribution, uncertainty propagation, or multiple-comparison correction is provided. Without such analysis, these comparisons should remain a minor speculative observation rather than a supporting validation of the model.

10. Spectral filtering and phase unwrapping require sensitivity analysis

The procedure applies a hard eigenvalue-magnitude cutoff, selects only one half of the complex plane, sorts eigenphases, performs phase unwrapping, and then linearly maps the resulting cumulative phase to Riemann-zero heights. Each of these choices can materially affect the output. The manuscript does not demonstrate invariance or stability with respect to cutoff selection, eigenvalue ordering, branch choice, perturbation of the matrix, or alternative unwrapping rules. Such sensitivity tests are essential before interpreting the resulting sequence as an intrinsic spectral signature.

11. Numerical sensitivity conflicts with the strength of the robustness claims

The Appendix reports that floating-point differences at approximately 10^{-15} can propagate through extremely long chaotic iterations and change the MSE by about a factor of two. This is a substantial sensitivity, not a negligible implementation detail. Results should therefore be reproduced across independent runs, compiler settings, arithmetic precision levels, grid resolutions, and Monte Carlo seeds, with error bars or confidence intervals. A single highly sensitive trajectory cannot support claims of robust topological correspondence.

12. Critical-point terminology and parameter interpretation are inconsistent

The manuscript distinguishes the usual Feigenbaum accumulation point near 1.401 from the empirically chosen baseline near 1.5437, but later refers to approximately 1.543689 as the “Feigenbaum critical point.” These are not interchangeable quantities. The dynamical meaning of the 1.543689 parameter must be stated consistently, and any connection between this value and the Riemann ground-state scale must be demonstrated rather than imposed by calibration.

Decisive Reasons for Rejection

1. The claimed “spectral isomorphism” is not mathematically established. The manuscript introduces a unitary operator U , but the numerical object actually diagonalized is a Markov/Ulam/Perron-Frobenius-type empirical transfer matrix. The relationship between these objects is not defined rigorously, and no self-adjoint/Hermitian operator with Riemann-zero ordinates as eigenvalues is constructed.
2. The numerical validation is circular because the target spectrum is used to tune the model. The discretization scale ε is selected by minimizing errors against the first Riemann zeros, while large- N parameters such as k_1 and k_2 are again optimized against the target spectrum. The reported agreement therefore cannot be interpreted as independent prediction.
3. The transfer-matrix construction and the non-autonomous spectral object are not adequately defined. A time average of instantaneous transition matrices is treated as if it represented the spectrum of the non-autonomous evolution, and the Monte Carlo expression given for P_{ij} is a joint transition frequency rather than, as written, a properly normalized conditional Markov transition probability.
4. The “vanishing intercept” in the conjugate-spectrum fit is likely imposed algebraically by symmetry. If the regression uses symmetric pairs (θ, γ) and $(-\theta, -\gamma)$, then the sample means vanish and an unconstrained least-squares intercept $b=0$ follows automatically. It should not be interpreted as a spontaneous physical calibration without demonstrating otherwise.
5. The manuscript contains material numerical inconsistencies. The k -values and MSE values reported in Table 1 differ substantially from those stated in Figure 4 and subsequently used in Figure 5. These are not rounding differences and prevent verification of which computation is authoritative.
6. The large- N fitting protocol is internally contradictory. The Results section describes single-point anchoring without global fitting, while the Appendix states that multi-point/global optimization is used for large datasets. This distinction is essential to the paper’s claims of predictive power.
7. The GUE comparison is asymmetric and does not test the established RMT correspondence. A globally scaled GUE spectrum is compared against the macroscopic counting-function trend, although the recognized GUE connection concerns unfolded local statistics. This baseline therefore does not constitute a decisive falsification test.
8. The robustness and statistical significance required for such a claim are missing. The paper reports extreme sensitivity to floating-point perturbations and makes post-hoc comparisons with experimental error peaks, but provides no systematic uncertainty budget, out-of-sample test, seed-to-seed variability, or statistical significance analysis.

The manuscript does not presently demonstrate a spectral isomorphism. Its central evidence is compromised by target-dependent calibration, insufficiently defined operator mathematics, potentially symmetry-forced regression behavior, contradictory fitting descriptions, and substantial internal numerical inconsistencies. These issues affect the validity of the principal conclusions and require a fundamentally redesigned study rather than a conventional revision.

Decision: REJECT